Physics from first principles

George M. Van Treeck ORCiD: 0000-0002-1729-2615

(Dated: 1 July 2026)

Euclidean volume and some distance equations are proved to be instances of ordered combinations (n-tuples) of elements from a list of disjoint sets. The linearly sequenceable and commutative properties of the sets of n-tuples and corresponding volume domain values are proved to limit a set of elements to at most 3 elements: for example, a set of 3 distance values, {width, height, depth}, and a set of 3 non-distance values, {time, mass, charge}, each value in R. Dividing the dimensions of Euclidean distance, time, mass, and charge into n number of fixed-sized intervals, for i in 1 to n, there are 3 direct and 3 inverse proportion ratios of the i-th Euclidean distance interval length to the i-th interval lengths of time, mass, and charge. Shorter and simpler derivations of well-known gravity, charge, electromagnetic, relativity, quantum physics equations and related constants from the ratios are presented. The inverse proportion ratios are used to add quantum extensions to the Schwarzschild metric, Newton's gravity force, and Coulomb's charge force equations, which predict that: 1) The gravity and charge constants do not exist where the quantum effects are measurable; 2) The force between two point masses and between two point charges peaks at the Planck unit length. The math proofs are verified in Rocq.

1

CONTENTS

Keywords: mathematical physics, gravity, charge, electromagnetism, relativity, quantum.

I. INTRODUCTION

A. Insufficient reduction

Much of physics relies on volume and distance equations. But mathematical analysis defines Euclidean volume as the product of interval lengths and defines the Euclidean distance equation, dot product, inner product, and metric space,[2,3] rather than deriving the equations from some set and number properties. Justification for the definitions is rigorous finger-pointing to Euclidean geometry, Cartesian geometry, and physics.

Further, the trend in physics has been increasingly complicated equations, assumptions, and mathematics, with increasing difficulty in finding solutions to the equations. Examples are the difficulty finding solutions to the Einstein field equations (general theory of relativity)[4,5] and Schrödenger's wave equation (quantum physics).[6]

And physics equations sometimes rely on assumptions that have not had a mathematical justification. For example, relativity and quantum theory both assume 3 dimensions of space. But integration, Hilbert spaces, linear algebra, (pseudo-)Riemann manifolds, and so on, have no restriction on the number of dimensions.[2,3,5,7,8]

The philosophy of reductionism posits that complex phenomena can be reduced to a few simple objects and relations. The first reduction step, in this article, proves that well-known volume and distance equations are reduced to (derived from) a few simple set and number properties. The second step is to identify three properties of the n-tuples and corresponding volume and distance equations salient to physics. The third reduction step is to show that many well-known physics equations, constants, and assumptions are reduced to (derived from) two consequences of those three properties.

B. First reduction step

If the underlying principles are simple, then the proofs that well-known geometry equations are reduced to (derived from) a few simple set relations and the properties of $\mathbb{R}$ should be simple, trivial even. But even apparently trivial proofs can have subtle flaws.

Therefore, each math proof, in this article, has a corresponding formal proof, which has been verified using using the Rocq proof verification system,[9] a verification tool used by mathematicians world-wide. The formal proofs are in the Rocq files, "euclidrelations.v" and "threed.v," which are included as ancillary files.

Let $|x_i|$ be the cardinal of (number of elements in) the abstract, countable set, $x_i \in \{x_1, \cdots, x_n\}$. And let the value, v_c, be the number of ordered combinations (n-tuples) of the elements of $x_1, \cdots, x_n$. Notionally:

$$x_i \in \{x_1, \cdots, x_n\}, \quad \bigcap_{i=1}^n x_i = \emptyset, \quad |x_i| \in \{0, \mathbb{N}\}, \quad v_c = \prod_{i=1}^n |x_i|. \tag{1}$$

Where each n-tuple corresponds to a scalar value, $\kappa \in \mathbb{R}$, it will be proved, without assuming the product of interval lengths, that:

$$v_c = \prod_{i=1}^n |x_i| \quad \Rightarrow \quad \exists\, v, s_i \in \mathbb{R}: \ \ v = \prod_{i=1}^n s_i. \tag{2}$$

For all $n > 1$, there are cases where different domain values, $|x_1|, \cdots, |x_n|$, multiplied yield the same range value, v_c. Inferring a domain value, d_c, from v_c, requires an inverse (bijective) function, $d_c = f_n^{-1}(v_c)$ and $v_c = f_n(d_c)$. The simplest bijective case for all n is the cuboid case, $|x_i| = d_c$. And, where v_c is the sum of the numbers of subset n-tuples, it will be proved that:

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \tag{3}$$

The $n = 2$ case is the dot product, where d is norm:

$$d^2 = \sum_{i=1}^{m}(\prod_{j=1}^{n=2} s_{i,j}) \ \wedge \ \exists\, a_i, b_i \in \mathbb{R}: \ \ s_{i,1} = a_i, \ \ s_{i,2} = b_i \ \ \Rightarrow \ \ d^2 = \sum_{i=1}^{m} a_i b_i \stackrel{\text{def}}{=} \mathbf{a}\cdot\mathbf{b}. \quad (4)$$

The inner product is where a_i and b_i are allowed to also be complex numbers, functions, and matrices.

And, where each summed volume, v_{c_i}, is also the bijective function, $d_{c_i}^n = v_{c_i}$:

$$d_c^n = v_c = \sum_{i=1}^{m} v_{c_i} = \sum_{i=1}^{m} d_{c_i}^n \quad \Rightarrow \quad d^n = \sum_{i=1}^{m} d_i^n. \quad (5)$$

$|d|$ is the p-norm (Minkowski distance),[10] which will be proved to imply the metric space properties.[3] The $n = 2$ case is the Euclidean distance.

C. Second reduction step

One property salient to physics: For all volumes, $v \in \mathbb{R}$, there exists $d \in \mathbb{R}$, which implies the dimension of all Euclidean distances $\subseteq \mathbb{R}$.

A second property salient to physics: All the set elements, $x_1, \cdots, x_n$ and volume domain values, $s_1, \cdots, s_n$, are linearly sequenceable via sequence functions, for example, sequenced via the successor and predecessor functions, $successor(x_i) = x_{i+1}$ and $predecessor(x_i) = x_{i-1}$.

A third property salient to physics: The commutative property of the union of subsets of n-tuples and the summing or multiplication operations defining the number n-tuples and corresponding Euclidean volume values (2) allow n number of values to be sequenced in all $n!$ permutations, for example, $v = width \times height \times depth = depth \times width \times height = \cdots$. The commutative property implies that each value has no intrinsic property that makes a value the first, second, ... , or last, which is a set of unordered values.

Reliably sequencing a set of unordered elements in the same linear order requires that the elements can be placed into a "list," which assigns a linear order for example, labeling the unordered elements as x_1, x_2, $\cdots x_n$. The linearly ordered list is sequenced via the successor and predecessor functions, $successor(x_i) = x_{i+1}$ and $predecessor(x_i) = x_{i-1}$. For example, assigning s_1 = width, s_2 = height, and s_3 = depth, the successor sequence, $v = \prod_{i=1}^{3} s_i$, reliably sequences the unordered set elements in the order, width$\rightarrow$height$\rightarrow$depth.

Only a cyclic list, where $successor(x_n) = x_1$ and $predecessor(x_1) = x_n$, allows each element to be the first element of some of those $n!$ permuted sequences. For example, where

$n = 3$, the volume cyclic successor sequences are: $v = s_1 \times s_2 \times s_3 = s_2 \times s_3 \times s_1 = s_3 \times s_1 \times s_2$, and the cyclic predecessor sequences are: $v = s_1 \times s_3 \times s_2 = s_2 \times s_1 \times s_3 = s_3 \times s_2 \times s_1$.

The only cyclic lists that allows linear sequencing in all $n!$ permutations, is where each list element is cyclic sequentially adjacent (either an *immediate* cyclic successor or an *immediate* cyclic predecessor) to every other element, referred to, in this article, as an immediate symmetric cyclic list (ISCL). For example, the cyclic permutations, $[x_2, x_3, x_1]$ and $[x_2, x_1, x_3]$, require x_1 and x_3 be cyclic sequentially adjacent. An ISCL will be proved to contain $n \leq 3$ elements.

D. Third reduction step

Two consequences of the Euclidean distance dimension $\subseteq \mathbb{R}$ property, the linearly sequenceable, and commutative properties of volume domain elements are:

1. **ISCL:** An ISCL of 3 distance dimensions, $\{r_1, r_2, r_3\}$, and an ISCL of 3 non-distance dimensions $\{t\ (time), m\ (mass),\ q\ (charge)\}$, each dimension having the $\subseteq \mathbb{R}$ membership property.

2. **Corresponding lengths:** Where the dimensions of Euclidean distance, time, mass, and charge are each divided into n number of fixed-sized interval lengths, for i in 1 to n, the i^{th} unit interval length, r_p, of Euclidean distance corresponds to the i^{th} unit interval lengths: t_p of time, m_p of mass, and q_p of charge, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

The correspondence of unit interval lengths implies 3 direct proportion unit ratios: $r/t = r_p/t_p$, $r/m = r_p/m_p$, and $r/q = r_p/q_p$, letting: $c_t = r_p/t_p$, $c_m = r_p/m_p$, and $c_q = r_p/q_p$. And the correspondence of unit interval lengths implies 3 inverse proportion unit ratios: $rt = t_p r_p$, $rm = m_p r_p$. and $rq = q_p r_p$, And let: $k_t = t_p r_p$, $k_m = m_p r_p$, and $k_q = q_p r_p$.

The gravity,[11,12] special relativity,[4,5] Schwarzschild black hole metric,[13,14] charge equations,[12], electromagnetism equations,[12] and related constants are derived from the 3 direct proportion unit ratios. The Planck-Einstein relation,[12] Compton wavelength,[12], Schrödenger wave equation,[6] Dirac wave equation,[15] and related constants are derived from combining some of the direct and inverse proportion unit ratios.

The inverse proportion ratios are used to add quantum extensions to Scharwzchild's metric, Newton's gravity force, and Coulomb's charge force, which predict: 1) The gravity (G) and charge (k_e) do not exist where the quantum effects are measurable; 2) The force between two point masses and between two point charges peak at the Planck unit length.

II. VOLUME

A. Euclidean volume

Theorem II.1. *Euclidean volume,*

$$\forall\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\},\ v_c = \prod_{i=1}^n |x_i| \ \Rightarrow\ v = \prod_{i=1}^n s_i, \quad s_i, v \in \mathbb{R}. \quad (6)$$

The formal proof, "Euclidean_volume," is in the Rocq file, euclidrelations.v.

Proof.

$$\forall v_c, |x_i| \in \{0, \mathbb{N}\},\ \ \kappa \in \mathbb{R}:\ \ v_c = \prod_{i=1}^n |x_i| \ \Leftrightarrow\ v_c\kappa = (\prod_{i=1}^n |x_i|)\kappa = \prod_{i=1}^n (|x_i|\kappa^{1/n}). \quad (7)$$

$$v_c\kappa = \prod_{i=1}^n (|x_i|\kappa^{1/n}) \ \wedge\ \exists\, v, s_i \in \mathbb{R}:\ \ v = v_c\kappa \ \wedge\ s_i = |x_i|\kappa^{1/n} \ \Rightarrow\ v = \prod_{i=1}^n s_i. \quad \square \quad (8)$$

Lemma II.2. *The number of n-tuples, v_c, is the sum of the number of n-tuples, v_{c_i}, in each disjoint subset of n-tuples, implies a volume is the sum of volumes,*

$$v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad v = \sum_{i=1}^m v_i, \quad v, v_i \in \mathbb{R}.$$

The formal proof, "sum_of _volumes," is in the Rocq file, euclidrelations.v.

Proof. From the condition of this theorem and the Euclidean volume theorem (II.1):

$$\forall\, \kappa \in \mathbb{R}:\ \ v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad v_c\kappa = (\sum_{i=1}^m v_{c_i})\kappa = \sum_{i=1}^m (v_{c_i}\kappa). \quad (9)$$

$$v_c\kappa = \sum_{i=1}^m (v_{c_i}\kappa) \quad \wedge \quad v = v_c\kappa \quad \wedge \quad v_i = v_{c_i}\kappa \quad \Rightarrow \quad v = \sum_{i=1}^m v_i. \quad \square \quad (10)$$

III. DISTANCE

Definition III.1. Bijective, countable domain value, d_c:

$$\exists\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\}, \quad v_c = \prod_{i=1}^n |x_i| = \prod_{i=1}^n d_c = d_c^n. \quad (11)$$

A. Vector inner product

Theorem III.2. *Sum of volumes distance:*

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i_j}).$$

The formal proof, "sum_of _volumes _distance," is in the Rocq file, euclidrelations.v.

Proof. Multiply both sides of the assumption by κ and apply the Euclidean volume proof (II.1) to v_i:

$$v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad v = \sum_{i=1}^m v_i, \quad v, v_i \in \mathbb{R}. \tag{12}$$

$$v = \sum_{i=1}^m v_i \quad \wedge \quad v_i = \prod_j^n s_{i,j} \quad \Rightarrow \quad v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \tag{13}$$

$$d_c^n = v_c \quad \Rightarrow \quad d_c^n \kappa = (d_c \kappa^{1/n})^n = v_c \kappa. \tag{14}$$

$$\exists\, d, v \in \mathbb{R} : d = d_c \kappa^{1/n} \quad \wedge \quad v = v_c \kappa \quad \wedge \quad (d_c \kappa^{1/n})^n = v_c \kappa \quad \Rightarrow \quad d^n = v. \tag{15}$$

$$d^n = v \quad \wedge \quad v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}) \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \quad \square \tag{16}$$

The $n = 2$ case is the dot product, where d is norm.:

$$d^2 = \sum_{i=1}^m (\prod_{j=1}^{n=2} s_{i,j}) \wedge \exists\, a_{i,1}, a_{i,2} \in \mathbb{R} : \; s_{i,1} = a_i, \; s_{i,2} = b_i \; \Rightarrow \; d^2 = \sum_{i=1}^m a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \tag{17}$$

The inner product is where a_i and b_i are allowed to also be complex numbers, functions, and matrices.

B. Minkowski distance (p-norm)

Theorem III.3. *Minkowski distance (p-norm):*

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} = \sum_{i=1}^m d_{c_i}^n \quad \Leftrightarrow \quad d^n = \sum_{i=1}^m d_i^n.$$

The formal proof, "Minkowski_distance," is in the Rocq file, euclidrelations.v.

Proof. From the sum of volumes distance theorem III.2:

$$d_{c_i}^n = v_{c_i} \quad \Rightarrow \quad d_i^n = v_i. \tag{18}$$

$$d^n = v = \sum_{i=1}^m v_i \quad \wedge \quad d_i^n = v_i \Rightarrow \quad d^n = \sum_{i=1}^m d_i^n \quad \square \tag{19}$$

C. Distance inequalities

Theorem III.4. *Distance triangle inequality*

$$\forall\, n \in \mathbb{N},\;\; v_a, v_b \geq 0:\;\; (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}.$$

The formal proof, distance_inequality, is in the Rocq file, euclidrelations.v.

Proof. Expand $(v_a^{1/n} + v_b^{1/n})^n$ using the binomial expansion:

$$\forall\, v_a, v_b \geq 0: \quad v_a + v_b \leq v_a + v_b + \textstyle\sum_{i=1}^{n}\binom{n}{k}(v_a^{1/n})^{n-k}(v_b^{1/n})^{k} + \textstyle\sum_{i=1}^{n}\binom{n}{k}(v_a^{1/n})^{k}(v_b^{1/n})^{n-k} = (v_a^{1/n} + v_b^{1/n})^n. \quad (20)$$

Take the n^{th} root of both sides of the inequality 20:

$$\forall\, v_a, v_b \geq 0,\;\; n \in \mathbb{N}: \quad v_a + v_b \leq (v_a^{1/n} + v_b^{1/n})^n \quad \Rightarrow \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}. \quad (21)$$

□

Theorem III.5. *Distance sum inequality*

$$\forall\, m, n \in \mathbb{N},\;\; a_i, b_i \geq 0: (\textstyle\sum_{i=1}^{m}(a_i^n + b_i^n))^{1/n} \leq (\textstyle\sum_{i=1}^{m} a_i^n)^{1/n} + (\textstyle\sum_{i=1}^{m} b_i^n)^{1/n}.$$

The formal proof, distance_sum_inequality, is in the Rocq file, euclidrelations.v.

Proof. Apply the distance triangle inequality (III.4):

$$\forall\, m, n \in \mathbb{N},\;\; v_a, v_b \geq 0:\;\; v_a = \textstyle\sum_{i=1}^{m} a_i^n \quad \wedge \quad v_b = \textstyle\sum_{i=1}^{m} b_i^n \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}$$
$$\Rightarrow \quad ((\textstyle\sum_{i=1}^{m} a_i^n) + (\textstyle\sum_{i=1}^{m} b_i^n))^{1/n} = (\textstyle\sum_{i=1}^{m}(a_i^n + b_i^n))^{1/n} \leq (\textstyle\sum_{i=1}^{m} a_i^n)^{1/n} + (\textstyle\sum_{i=1}^{m} b_i^n)^{1/n}. \quad \square \quad (22)$$

D. Metric Space

All Minkowski distances (p-norms) imply the metric space properties. The formal proofs: triangle_inequality, symmetry, non_negativity, and identity_of_indiscernibles are in the Rocq file, euclidrelations.v.

Theorem III.6. *Triangle Inequality:*

$$d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$$
$$\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w).$$

Proof. $\forall\, p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$:

$$(u^p + w^p)^{1/p} \leq ((u^p + w^p) + 2v^p)^{1/p} = ((u^p + v^p) + (v^p + w^p))^{1/p}. \tag{23}$$

Apply the distance triangle inequality (III.4) to the inequality 23:

$$\begin{aligned}
&(u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n} \\
&\qquad \wedge \quad v_a = u^p + v^p \quad \wedge \quad v_b = v^p + w^p \\
&\Rightarrow \quad (u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} \\
&\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w). \quad \square
\end{aligned} \tag{24}$$

Theorem III.7. *Symmetry:*
$d(s_1, s_2) = (\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad v = s_2 \quad \Rightarrow \quad d(u, v) = d(v, u).$

Proof. By the commutative law of addition:

$$\begin{aligned}
&\forall\, p \,:\, p \geq 1, \quad d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p} = (s_1^p + s_2^p)^{1/p} \\
&\qquad \Rightarrow \quad d(u, v) = (u^p + v^p)^{1/p} = (v^p + u^p)^{1/p} = d(v, u). \quad \square
\end{aligned} \tag{25}$$

Theorem III.8. *Non-negativity:*

$$d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \in N \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(u, v) \geq 0.$$

Proof.

$$\forall\, n \geq 1 \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(s_1, s_2) = (s_1^p + s_1^p)^{1/p} \geq 0. \tag{26}$$

$$\forall\, u = s_1,\ v = s_2 \ \wedge\ d(s_1, s_2) = (s_1^p + s_1^p)^{1/p} \geq 0 \ \Rightarrow\ d(u, v) = (u^p + v^p)^{1/p} \geq 0. \quad \square \tag{27}$$

Theorem III.9. *Identity of Indiscernibles:* $d(u, u) = 0$.

Proof. From the non-negativity property (III.8):

$$d(u, w) \geq 0 \quad \wedge \quad d(u, v) \geq 0 \quad \wedge \quad d(v, w) \geq 0 \tag{28}$$

$$\Rightarrow \quad \exists\, d(u, w) = d(u, v) = d(v, w) = 0. \tag{29}$$

$$d(u, w) = d(v, w) = 0 \quad \Rightarrow \quad u = v. \tag{30}$$

$$d(u, v) = 0 \quad \wedge \quad u = v \quad \Rightarrow \quad d(u, u) = 0. \quad \square \tag{31}$$

IV. PROPERTIES LIMITING A SET TO AT MOST 3 ELEMENTS

The following definitions and proof use first order logic. A Horn clause-like expression is used, here, to make the proof easier to read. By convention, the proof goal, in a Horn clause, is on the left side of the implication sign ($\leftarrow$) and supporting facts are on the right side. The formal proofs in the Rocq file threed.v are: adj111, adj122, adj212, adj123, adj133, adj233, adj213, adj313, adj323, and not_all_mutually_adjacent_gt_3.

Definition IV.1. Immediate Cyclic Successor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$

$$Successor(m, n, setsize) \quad \leftarrow \quad (m = setsize \wedge n = 1) \quad \vee \quad (n = m + 1 \leq setsize). \quad (32)$$

Definition IV.2. Immediate Cyclic Predecessor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$

$$Predecessor(m, n, setsize) \quad \leftarrow \quad (m = 1 \wedge n = setsize) \quad \vee \quad (n = m - 1 \geq 1). \quad (33)$$

Definition IV.3. Adjacent: Element m is sequentially adjacent to element n if the immediate cyclic successor of m is n or the immediate cyclic predecessor of m is n. Notionally:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$

$$Adjacent(m, n, setsize) \leftarrow Successor(m, n, setsize) \;\vee\; Predecessor(m, n, setsize). \quad (34)$$

Definition IV.4. Immediate Symmetric Cylic List (ISCL), where every set element is sequentially adjacent to every other element):

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} : \quad Adjacent(m, n, setsize). \quad (35)$$

Theorem IV.5. *An ISCL is limited to at most 3 elements.*

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} : \quad Adjacent(m, n, setsize) \quad \Rightarrow \quad setsize \leq 3. \quad (36)$$

Proof.

Every element is adjacent to every other element, where $setsize \in \{1,\ 2,\ 3\}$:

$$Adjacent(1,1,1) \leftarrow Successor(1,1,1) \leftarrow (m = setsize \wedge n = 1). \tag{37}$$

$$Adjacent(1,2,2) \leftarrow Successor(1,2,2) \leftarrow (n = m + 1 \leq setsize). \tag{38}$$

$$Adjacent(1,2,3) \leftarrow Successor(1,2,3) \leftarrow (n = m + 1 \leq setsize). \tag{39}$$

$$Adjacent(2,1,3) \leftarrow Predecessor(2,1,3) \leftarrow (n = m - 1 \geq 1). \tag{40}$$

$$Adjacent(3,1,3) \leftarrow Successor(3,1,3) \leftarrow (n = setsize \wedge m = 1). \tag{41}$$

$$Adjacent(1,3,3) \leftarrow Predecessor(1,3,3) \leftarrow (m = 1 \wedge n = setsize). \tag{42}$$

$$Adjacent(2,3,3) \leftarrow Successor(2,3,3) \leftarrow (n = m + 1 \leq setsize). \tag{43}$$

$$Adjacent(3,2,3) \leftarrow Predecessor(3,2,3) \leftarrow (n = m - 1 \geq 1). \tag{44}$$

Element 2 is the only immediate successor of element 1 for all $setsize \geq 3$, which implies element 3 is not ($\neg$) an immediate successor of element 1 for all $setsize \geq 3$:

$$\begin{multline}\neg Successor(1,3,setsize \geq 3) \leftarrow \\ Successor(1,2,setsize \geq 3) \leftarrow (n = m + 1 \leq setsize).\end{multline} \tag{45}$$

Element $n = setsize > 3$ is the only immediate predecessor of element 1, which implies element 3 is not ($\neg$) an immediate predecessor of element 1 for all $setsize > 3$:

$$\begin{multline}\neg Predecessor(1,3,setsize \geq 3) \leftarrow Predecessor(1,setsize,setsize > 3) \leftarrow \\ (m = 1 \wedge n = setsize > 3).\end{multline} \tag{46}$$

For all $setsize > 3$, some elements are not ($\neg$) sequentially adjacent to every other element (not immediate symmetric):

$$\begin{multline}\neg Adjacent(1,3,setsize > 3) \leftarrow \\ \neg Successor(1,3,setsize > 3) \quad \wedge \quad \neg Predecessor(1,3,setsize > 3). \quad \square\end{multline} \tag{47}$$

The Symmetric goal matches the Adjacent goals 37 and fails for all "setsize" greater than three.

V. DERIVATION OF PHYSICS EQUATIONS AND CONSTANTS

A. The direct and inverse unit ratios

Application of the ISCL property of volume (IV.5) and the Euclidean distance dimension $\subseteq \mathbb{R}$ (III.3):

1. **ISCL:** $\{r_1, r_2, r_3\}$ is an ISCL of 3 "distance" dimensions, each dimension having the $\subseteq \mathbb{R}$ set membership property and $\{t\ (time), m\ (mass),\ q\ (charge)\}$ is an ISCL of 3 "non-distance" dimensions, each dimension $\subseteq \mathbb{R}$. There is a total of 6 dimensions with the $\subseteq \mathbb{R}$ property: $r_1,\ r_2,\ r_3,\ t,\ m,\ q$.

2. **Corresponding lengths:** Where the dimensions of Euclidean distance, time, mass, and charge are each divided into n number of fixed-sized interval lengths, for i in 1 to n, the i^{th} unit interval length, r_p, of Euclidean distance corresponds to the i^{th} unit interval lengths: t_p of time, m_p of mass, and q_p of charge, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

There are 3 direct proportion ratios from the properties of $\mathbb{R}$:

$$\forall\, r_p, t_p, m_p, q_p, k_1, k_2, k_3 \in \mathbb{R},\ \exists\, r, t, m, q \in \mathbb{R}:$$

$$\frac{r_p}{t_p} = \frac{r_p k_1}{t_p k_1} = \frac{r}{t} \quad \wedge \quad \frac{r_p}{m_p} = \frac{r_p k_2}{m_p k_2} = \frac{r}{m} \quad \wedge \quad \frac{r_p}{q_p} = \frac{r_p k_3}{q_p k_3} = \frac{r}{q}. \tag{48}$$

$$\exists\, c, c_t, c_m, c_q \in \mathbb{R}: \quad r_p/t_p = c_t = c \quad \wedge \quad r_p/m_p = c_m \quad \wedge \quad r_p/q_p = c_q$$

$$\Rightarrow \quad r/t = r_p/t_p = c_t = c \quad \wedge \quad r/m = r_p/m_p = c_m \quad \wedge \quad r/q = r_p/q_p. = c_q. \tag{49}$$

And there are 3 inverse proportion ratios from the properties of $\mathbb{R}$:

$$\forall\, r_p, t_p, m_p, q_p, k_1, k_2, k_3 \in \mathbb{R},\ \exists\, r, t, m, q \in \mathbb{R}:$$

$$r_p t_p = r_p t_p \left(\frac{k_1}{k_1}\right) = \left(\frac{r_p}{k_1}\right)(t_p k_1) = rt \quad \wedge \quad r_p m_p = r_p m_p \left(\frac{k_2}{k_2}\right) = \left(\frac{r_p}{k_2}\right)(m_p k_2) = rm$$

$$\wedge \quad r_p q_p = r_p q_p \left(\frac{k_3}{k_3}\right) = \left(\frac{r_p}{k_3}\right)(q_p k_3) = rq. \tag{50}$$

$$\exists\, k_t, k_m, k_q \in \mathbb{R}: \quad r_p t_p = k_t \quad \wedge \quad r_p m_p = k_m \quad \wedge \quad r_p q_p = k_q$$

$$\Rightarrow \quad rt = t_p r_p = k_t \quad \wedge \quad rm = m_p r_p = k_m \quad \wedge \quad rq = q_p r_p = k_q. \tag{51}$$

B. G, Newton, Gauss, Poisson gravity laws

From equations 49, linear acceleration, r/t^2, can be related to mass, m, and distance, r:

$$r = c_m m \ \wedge \ r = c_t t \ \Rightarrow \ \frac{r}{(c_t t)^2} = \frac{c_m m}{r^2} \ \Rightarrow \ \frac{r}{t^2} = \frac{(c_m c_t^2) m}{r^2} = \frac{Gm}{r^2}, \tag{52}$$

where $G = c_m c_t^2 = r_p^3/m_p t_p^2$ conforms to the SI units: $m^3 \cdot kg^{-1} \cdot s^{-2}$.[11]

Newton's law follows from multiplying both sides of equation 52 by m:

$$\frac{r}{t^2} = \frac{Gm}{r^2} \ \Leftrightarrow \ F \overset{\text{def}}{=} \frac{mr}{t^2} = \frac{Gm^2}{r^2}. \tag{53}$$

$$F = \frac{Gm^2}{r^2} \ \wedge \ \forall\, m \in \mathbb{R} : \exists\, m_1, m_2 \in \mathbb{R} : \ m_1 m_2 = m^2 \quad \Rightarrow \quad F = \frac{G m_1 m_2}{r^2}. \tag{54}$$

A new interpretation of Gauss's and Poisson's gravity laws is presented here. Equation 52 relates linear acceleration, r/t^2, to mass and distance. Gauss's gravity field, $\mathbf{g}$, and Poisson's gravity field, $-\nabla\Phi(\tilde{\mathbf{r}}, t)$, is the angular acceleration, $2\pi r/t^2$. Applying equation 52:

$$\frac{r}{t^2} = -\frac{Gm}{|\tilde{\mathbf{r}}|^2} \ \wedge \ \mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = \frac{2\pi r}{t^2} \ \Rightarrow \ \mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = -\frac{2\pi G m}{|\tilde{\mathbf{r}}|^2}. \tag{55}$$

$$\mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = -\frac{2\pi G m}{|\tilde{\mathbf{r}}|^2} \quad \Rightarrow \quad \nabla \cdot \mathbf{g} = \nabla^2\Phi(\tilde{\mathbf{r}}, t) = \frac{4\pi G m}{|\tilde{\mathbf{r}}|^3} = 4\pi G\rho, \tag{56}$$

where $\rho = m/|\tilde{\mathbf{r}}|^3$ is the mass density:

C. Special relativity

$$\forall\, \tau \in \{t, m, q\},\ r^2 = r'^2 + r_v^2,\ \exists\, \mu, \nu : \ r = \mu\tau \quad \wedge \quad r_v = \nu\tau \quad \Rightarrow \quad (\mu\tau)^2 = r'^2 + (\nu\tau)^2$$
$$\Rightarrow \quad r' = \sqrt{(\mu\tau)^2 - (\nu\tau)^2} = \mu\tau\sqrt{1 - (\nu/\mu)^2}. \tag{57}$$

Local frame distance, r', contracts relative to a distant observer frame distance, r, as $\nu \to \mu$:

$$r' = \mu\tau\sqrt{1 - (\nu/\mu)^2} \quad \wedge \quad \mu\tau = r \quad \Rightarrow \quad r' = r\sqrt{1 - (\nu/\mu)^2}. \tag{58}$$

A distant observer frame type, τ, dilates relative to the local observer frame type, τ', as $\nu \to \mu$:

$$\mu\tau = r'/\sqrt{1 - (\nu/\mu)^2} \quad \wedge \quad r' = \mu\tau \quad \Rightarrow \quad \tau = \tau'/\sqrt{1 - (\nu/\mu)^2}. \tag{59}$$

Where τ is type, time, the space-like flat Minkowski spacetime event interval is:

$$\mathrm{d}r^2 = \mathrm{d}r'^2 + \mathrm{d}r_v^2 \quad \wedge \quad \mathrm{d}r_v^2 = \mathrm{d}r_1^2 + \mathrm{d}r_2^2 + \mathrm{d}r_3^2 \quad \wedge \quad \mathrm{d}(\mu\tau) = \mathrm{d}r$$
$$\Rightarrow \ \mathrm{d}r'^2 = \mathrm{d}(\mu\tau)^2 - \mathrm{d}r_1^2 - \mathrm{d}r_2^2 - \mathrm{d}r_3^2. \tag{60}$$

D. Analytic general relativity solutions

The Einstein field equation (EFE) is:

$$\mathbf{G}_{\mu,\nu} + \Lambda g_{\mu,\nu} = \kappa T_{\mu,\nu}, \quad \mathbf{G}_{\mu,\nu} = \mathbf{R} + \frac{1}{2} R g_{\mu,\nu}. \tag{61}$$

where: $\mathbf{G}_{\mu,\nu}$ is Einstein's tensor, $\mathbf{R}$ is the Ricci curvature, R is the scalar curvature, $g_{\mu,\nu}$ is the metric tensor, Λ is a cosmological constant, $\kappa = 8\pi G/c^4$, is Einstein's constant, and $T_{\mu,\nu}$ is the stress-energy-momentum tensor.[4,5]

The EFE extends the previously derived Poisson's law of gravity (56) to include a time component. The Minkowski spacetime interval component for time is $c^2 g_{0,0} \mathrm{d}t^2$:

$$\frac{c^2}{2}(\mathbf{G}_{0,0} + \Lambda g_{0,0}) = \nabla^2 \Phi(\tilde{\mathbf{r}}, t) = -4\pi G \rho = -4\pi G \frac{m}{|\mathbf{r}|^3} = -\frac{4\pi G}{c^2} \cdot \frac{mc^2}{|\mathbf{r}|^3} = \frac{4\pi G}{c^2} T_{0,0}. \tag{62}$$

$$\frac{c^2}{2}(\mathbf{G}_{0,0} + \Lambda g_{0,0}) = \frac{4\pi G}{c^2} T_{0,0} \quad \Rightarrow \quad \mathbf{G}_{0,0} + \Lambda g_{0,0} = \frac{8\pi G}{c^4} T_{0,0} = \kappa T_{0,0}. \tag{63}$$

Note: 1) The laws of physics determine the amount of spacetime curvature and the distribution of stress, energy, and momentum. 2) The laws of physics are specified as the components of the metric, $g_{\mu,\nu}$. Derivation of the Schwarzschild metric, in next subsection, is an example of the following simple steps to derive exact (analytic) metrics independent of the EFE:

Step 1) Use the unit ratios and equations (laws) derived from the ratios, for example, the ratio-derived special relativity equations (58) and the ratio-derived constant, G (52), to derive the scalar functions (laws) assigned to the components of $g_{\mu,\nu}$.

Step 2) The 4D flat spacetime interval equation is an instance of the 2D equation (60). Therefore, derive the metric first as a 2D metric and expand later to a 4D metric.

(Optional step 2d) The 2D metric tensor allows uses the much simpler 2D Ricci curvature (which, in 2D, is the Gaussian curvature). And the 2D tensors reduce the number of independent equations to solve. The 2D solutions can next be used to set constraints on the solutions in the 4D tensor, much like an Euclidean distance value places constraints on the coordinate distance values.

Step 3) Translate from 2D to 4D. For spherically symmetric cases, the simplest method to translate from 2D to 4D is to use spherical coordinates, where the first 2 diagonal components of $g_{\nu,\mu}$ remain unchanged and the other 2 diagonal components are functions of the longitude angle, θ, and latitude angle, ϕ.

E. Schwarzschild's time dilation and metric

From equations 58 and 49:

$$r = c_m m \quad \Rightarrow \quad c_m m/r = 1 \quad \Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2}. \tag{64}$$

Where v_{escape} is the escape velocity and KE is the kinetic energy:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv_{escape}^2$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v_{escape}^2/rc^2}. \tag{65}$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v_{escape}^2/rc^2}. \quad \wedge \quad v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m c^2/rc^2}. \tag{66}$$

Combining equation 66 with the derivation of G (52):

$$c_m c^2 = G \wedge \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m c^2/rc^2} \Rightarrow \sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2}. \tag{67}$$

Combining equation 67 with equation 59 yields Schwarzschild's gravitational time dilation:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2} \wedge t' = t\sqrt{1-(v^2/c^2)} \Rightarrow t' = t\sqrt{1-2Gm/rc^2}. \tag{68}$$

From equations 58 and 67, where Schwarzschild defined the black hole event horizon radius as $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$.[13,14]

$$r' = r\sqrt{1-(v/c)^2} = r\sqrt{1-2Gm/rc^2} \quad \wedge \quad \alpha \stackrel{\text{def}}{=} 2Gm/c^2 \quad \Rightarrow \quad r' = r\sqrt{1-\alpha/r}. \tag{69}$$

Applying equation 69 to the time-like spacetime interval equation 60, where $\mathrm{d}s^2 < 0$:

$$r' = r\sqrt{1-\alpha/r} \quad \wedge \quad \mathrm{d}s^2 = \mathrm{d}r'^2 - \mathrm{d}r^2$$
$$\Rightarrow \quad \mathrm{d}s^2 = (\sqrt{1-\alpha/r}\mathrm{d}r)^2 - (\mathrm{d}r'/\sqrt{1-\alpha/r})^2 = (1-\alpha/r)\mathrm{d}r^2 - (1-\alpha/r)^{-1}\mathrm{d}r'^2. \tag{70}$$
$$\mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}r^2 - (1-\alpha/r)^{-1}\mathrm{d}r'^2 \quad \wedge \quad \mathrm{d}r = \mathrm{d}(ct) \quad \wedge \quad c = 1 \quad \wedge$$
$$\lim_{\mathrm{d}s\to 0} \mathrm{d}r' = \lim_{\mathrm{d}s\to 0} \mathrm{d}r \quad \Rightarrow \quad \lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2. \tag{71}$$

Using spherical coordinates to translate from 2D to 4D yields the $+ - --$ form of Schwarzschild's black hole metric:[13,14]

$$\lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2$$
$$\Rightarrow \quad \lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + sin^2\theta \mathrm{d}\phi^2)$$
$$\Rightarrow \quad g_{\mu,\nu} = diag[(1-\alpha/r),\ -(1-\alpha/r)^{-1},\ -r^2,\ -r^2 sin^2\theta]. \quad (72)$$

F. k_e, ε_0, E, Coulomb and Gauss laws

From equations 49, linear acceleration, r/t^2, can be related to charge, q, and distance, r:

$$r = c_q q \ \wedge \ r = c_t t \ \Rightarrow \ \frac{r}{(c_t t)^2} = \frac{c_q q}{r^2} \ \Rightarrow \ \frac{r}{t^2} = \frac{(c_q c_t^2)q}{r^2}. \quad (73)$$

$$r = c_m m \quad \wedge \quad r = c_q q \quad \Rightarrow \quad c_m m = c_q q \quad \Rightarrow \quad m = (c_q/c_m)q. \quad (74)$$

Combining equations 73 and 74 yields Coulombs's law:

$$\frac{r}{t^2} = \frac{(c_q c_t^2)q}{r^2} \ \wedge \ m = (c_q/c_m)q \ \Rightarrow \ F \stackrel{\text{def}}{=} \frac{mr}{t^2} = \frac{(c_{q^2} c_t^2/c_m)q}{r^2} = \frac{k_e q^2}{r^2}. \quad (75)$$

$k_e = c_q^2 c_t^2/c_m = m_p r_p^3/t_p^2 q_p^2$, conforms to the SI units: $kg \cdot m^3 \cdot s^{-2} \cdot C^{-2} \stackrel{\text{def}}{=} N \cdot m^2 \cdot C^{-2}$.[12]

$$\forall\, q \in \mathbb{R}\ \exists\, q_1, q_2 \in \mathbb{R}:\ q_1 q_2 = q^2 \quad \wedge \quad F = \frac{k_e q^2}{r^2}. \quad \Rightarrow \quad F = \frac{k_e q_1 q_2}{r^2}.. \quad (76)$$

A new interpretation of Gauss's electric field law is presented here. Equation 73 relates the linear acceleration, r/t^2, to charge and distance. Gauss's electric field, **E**, relates angular acceleration, $2\pi r/t^2$ to charge and distance. Applying equation 75:

$$F_C = \frac{mr}{t^2} = \frac{k_e q^2}{r^2} \quad \Rightarrow \quad \exists\, F_E \in \mathbb{R}:\ F_E = \frac{2\pi m r}{t^2} = \frac{2\pi k_e q^2}{r^2}. \quad (77)$$
$$F_E = \frac{2\pi k_e q^2}{r^2} = q(\frac{2\pi k_e q}{r^2}) \quad \wedge \quad \exists\, E \in R:\ E = \frac{2\pi k_e q}{r^2} \quad \Rightarrow \quad F_E = qE. \quad (78)$$

The electric field, $E = 2\pi k_e q/r^2$, conforms to the SI units $kg \cdot m \cdot s^{-2} \cdot C^{-1} \stackrel{\text{def}}{=} kg \cdot m \cdot s^{-3} \cdot \cdot A^{-1}$.

$$\mathbf{E} = \frac{2\pi k_e q}{|\vec{\mathbf{r}}|^2} \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3}. \quad (79)$$

$$\nabla \cdot \mathbf{E} = -\frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3} \quad \wedge \quad \varepsilon_0 \stackrel{\text{def}}{=} \frac{1}{4\pi k_e} \quad \wedge \quad \rho \stackrel{\text{def}}{=} \frac{q}{|\vec{\mathbf{r}}|^3} \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\rho/\varepsilon_0, \quad (80)$$

which is Gauss's electric field law in differential form.[12]

G. B, μ_0, and Lorentz's law

Applying the distance contraction equation, 58, to equation 78, where r' is the distant observer frame of reference and r is local (rest) frame of reference:

$$r' = r/\sqrt{1 - v^2/c^2} \quad \wedge \quad F' = \frac{2\pi k_e q^2}{r'^2} \quad \Rightarrow \quad F = \frac{2\pi k_e q^2(1 - v^2/c^2)}{r^2}. \tag{81}$$

From equation 78:

$$E = \frac{2\pi k_e q}{r^2} \;\wedge\; F = \frac{2\pi k_e q^2(1 - v^2/c^2)}{r^2} \;\Rightarrow\; F = q(E - \frac{(2\pi k_e/c^2)q}{r^2}v^2). \tag{82}$$

$$F = q(E - \frac{(2\pi k_e/c^2)q}{r^2}v^2) \;\wedge\; \exists\, B:\; B = \frac{(2\pi k_e/c^2)vq}{r^2} \;\Rightarrow\; F = q(E - Bv), \tag{83}$$

where the magnetic field, $B = (2\pi k_e/c)q/r^2$, conforms to the base SI units: $kg \cdot C^{-1} \cdot s^{-1}$. $\stackrel{\text{def}}{=}$ $kg \cdot A^{-1} \cdot s^{-2}$ $\stackrel{\text{def}}{=}$ T. And assigning the constant portion to μ_0:

$$B = \frac{(2\pi k_e/c^2)vq}{r^2} \quad \Rightarrow \quad \frac{\partial B}{\partial r} = \frac{(4\pi k_e/c)v^2 q}{r^2} = \frac{\mu_0 vq}{r^3}. \tag{84}$$

where $\mu_0 = 4\pi k_e/c^2$ conforms to the SI units $kg \cdot m \cdot C^{-2} \stackrel{\text{def}}{=} kg \cdot m \cdot s^{-2} A^{-2} \stackrel{\text{def}}{=} N \cdot A^{-2}$.

Translating equation 83 to vector form:

$$F = q(E - Bv) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \vec{\mathbf{v}}). \tag{85}$$

$$\mathbf{B} \times \vec{\mathbf{v}} = -(\vec{\mathbf{v}} \times \mathbf{B})) \quad \wedge \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \vec{\mathbf{v}}) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} + \vec{\mathbf{v}} \times \mathbf{B}), \tag{86}$$

which is Lorentz law in the local frame of reference.

H. Faraday's law

From the magnetic field equation 86, where the electric and magnetic fields are propagating at the speed, $v = c$:

$$B = \frac{(2\pi k_e/c^2)vq}{r^2} \quad \wedge \quad v = c \quad \wedge \quad r = ct \quad \Rightarrow \quad B = \frac{(2\pi k_e/c^3)q}{t^2}. \tag{87}$$

$$B = \frac{(2\pi k_e/c^3)q}{t^2} \quad \Rightarrow \quad \frac{\partial B}{\partial t} = -\frac{(4\pi k_e/c^3)q}{t^3}. \tag{88}$$

$$\frac{\partial B}{\partial t} = -\frac{(4\pi k_e/c^3)q}{t^3} \quad \wedge \quad r = ct \quad \Rightarrow \quad \frac{\partial B}{\partial t} = -\frac{4\pi k_e q}{r^3}. \tag{89}$$

Faraday's law is the case, where the electric field, $\mathbf{E}$, is a single radial vector, $\mathbf{E}_{\tilde{\mathbf{r}}_1}$ acting on some point, P. Therefore, at point, P, $\mathbf{E}_{\tilde{\mathbf{r}}_2} = 0$ and $\mathbf{E}_{\tilde{\mathbf{r}}_3} = 0$. From the Lorentz equation

86, the magnetic field, $\mathbf{B}$ at point P, is perpendicular to $\mathbf{E}_{\tilde{\mathbf{r}}_1}$ (aligned on $\tilde{\mathbf{r}}_2$). For a constant force, the Lorentz equation makes the electric field, $\mathbf{E}_{\tilde{\mathbf{r}}_1}$, a function of $\mathbf{B}_{\tilde{\mathbf{r}}_2}$. Combining with equation 79:

$$\mathbf{E}_{\tilde{\mathbf{r}}_1} = 2\pi k_e q/|\tilde{\mathbf{r}}_2|^2 = f(\mathbf{B}_{\tilde{\mathbf{r}}_2}) \quad \wedge \quad \mathbf{E}_{\tilde{\mathbf{r}}_2} = 0 \quad \wedge \quad \mathbf{E}_{\tilde{\mathbf{r}}_3} = 0$$

$$\Rightarrow \quad \nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial \tilde{\mathbf{r}}_1^2} & \frac{\partial}{\partial \tilde{\mathbf{r}}_2^2} & \frac{\partial}{\partial \tilde{\mathbf{r}}_3^2} \\ \mathbf{E}_{\tilde{\mathbf{r}}_1} & 0 & 0 \end{vmatrix} = (\frac{\partial \mathbf{E}_{r_3}}{\partial \tilde{\mathbf{r}}_2^2} - \frac{\partial \mathbf{E}_{r_2}}{\partial \tilde{\mathbf{r}}_3^2})\hat{\mathbf{i}} + (\frac{\partial \mathbf{E}_{r_1}}{\partial \tilde{\mathbf{r}}_3^2} - \frac{\partial \mathbf{E}_{r_3}}{\partial \tilde{\mathbf{r}}_1^2})\hat{\mathbf{j}} + (\frac{\partial \mathbf{E}_{r_2}}{\partial \tilde{\mathbf{r}}_1^2} - \frac{\partial \mathbf{E}_{r_1}}{\partial \tilde{\mathbf{r}}_2^2})\hat{\mathbf{k}}$$

$$= 0\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + (0 - \frac{\partial 2\pi k_e q/|\tilde{\mathbf{r}}_2|^2}{\partial |\tilde{\mathbf{r}}_2|^2})\hat{\mathbf{k}} \quad = \quad \frac{4\pi k_e q}{|\tilde{\mathbf{r}}_2|^3}\hat{\mathbf{k}} \quad \equiv \quad \nabla \times \mathbf{E} = \frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3}. \tag{90}$$

Combining equations 90 and 89 yields Maxwell-Faraday's law:[12]

$$\nabla \times \mathbf{E} = \frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3} \quad \wedge \quad \frac{\partial \mathbf{B}}{\partial t} = -\frac{4\pi k_e q}{|\vec{\mathbf{r}}|^3} \quad \Rightarrow \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{91}$$

I. Maxwell's wave equations

From the Gauss's electric field equations 80 and 84:

$$\mathbf{E} = \frac{2\pi k_e q}{|\vec{\mathbf{r}}|^2} \quad \Rightarrow \quad \nabla^2 \mathbf{E} = \frac{12\pi k_e q}{|\vec{\mathbf{r}}|^4}. \tag{92}$$

$$\mathbf{E} = \frac{2\pi k_e q}{|\vec{\mathbf{r}}|^2} \quad \wedge \quad r = ct \quad \Rightarrow \quad \mathbf{E} = \frac{2\pi k_e q}{(ct)^2} \quad \Rightarrow \quad \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4}. \tag{93}$$

$$\frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4} \quad \wedge \quad r = ct \quad \Rightarrow \quad \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\vec{\mathbf{r}}|^4}. \tag{94}$$

$$\nabla^2 \mathbf{E} = \frac{12\pi k_e q}{|\vec{\mathbf{r}}|^4} \quad \wedge \quad \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\vec{\mathbf{r}}|^4} \quad \Rightarrow \quad \nabla^2 \mathbf{E} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}. \tag{95}$$

$$\nabla^2 \mathbf{E} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \wedge \quad \varepsilon_0 \mu_0 = \frac{1}{4\pi k_e} \cdot \frac{4\pi k_e}{c^2} = \frac{1}{c^2} \quad \Rightarrow \quad \nabla^2 \mathbf{E} = \varepsilon_0 \mu_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \tag{96}$$

which is Maxwell's electric field wave equation. And from equation (89):

$$\frac{\partial B}{\partial t} = -\frac{4\pi k_e q}{r^3} \quad \wedge \quad r = |\vec{\mathbf{r}}| \quad \Rightarrow \quad \nabla^2 \mathbf{B} = \frac{12\pi k_e q}{|\vec{\mathbf{r}}|^4}. \tag{97}$$

$$\mathbf{B} = \frac{2\pi k_e q}{|\vec{\mathbf{r}}|^2} \quad \wedge \quad r = ct \quad \Rightarrow \quad \mathbf{B} = \frac{2\pi k_e q}{(ct)^2} \quad \Rightarrow \quad \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4}. \tag{98}$$

$$\frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4} \quad \wedge \quad r = ct \quad \Rightarrow \quad \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\vec{\mathbf{r}}|^4}. \tag{99}$$

$$\nabla^2 \mathbf{B} = -12\pi k_e q/|\vec{\mathbf{r}}|^4 \quad \wedge \quad \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\vec{\mathbf{r}}|^4} \quad \Rightarrow \quad \nabla^2 \mathbf{B} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}. \tag{100}$$

$$\nabla^2 \mathbf{B} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \wedge \quad \varepsilon_0 \mu_0 = \frac{1}{4\pi k_e} \cdot \frac{4\pi k_e}{c^2} = \frac{1}{c^2} \quad \Rightarrow \quad \nabla^2 \mathbf{B} = \varepsilon_0 \mu_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}, \tag{101}$$

which is Maxwell's magnetic field wave equation.

J. The fine structure constants, α and α_G

New, simpler motivation for the fine structure constants is presented here. Each type of fine structure is the ratio of an electron's particle's static field force to the force of a wave caused by the motion of an electron, which reduces to the ratio of the square of the units forces:

$$\alpha_\tau = \frac{F_{\tau_e}}{F_{\tau_p}} = \frac{K_\tau \tau_e^2/r^2}{K_\tau \tau_p^2/r^2} = \frac{\tau_e^2}{\tau_p^2}, \quad \tau \in \{m, q\}. \tag{102}$$

For example, the fine structure electron charge coupling constant is the ratio of stationary and moving Coulomb charge (electromagnetic) wave forces, which simplifies to:

$$\alpha = \frac{k_e q_e^2/r^2}{k_e q_p^2/r^2} = \frac{q_e^2}{q_p^2} \approx 0.0072973526, \tag{103}$$

where q_e is the elementary (electron) charge and q_p is the Planck charge unit.

The fine structure electron gravity coupling constant, α_G, is the ratio of stationary to moving Newton gravity wave forces, which simplifies to:

$$\alpha_G = \frac{G m_e^2/r^2}{G m_p^2/r^2} = \frac{m_e^2}{m_p^2} \approx 1.751809 \cdot 10^{-45}, \tag{104}$$

where m_e is the electron mass and m_p is the Planck mass unit.

K. $\hbar$, h, and Planck-Einstein relation

Applying both the direct proportion ratio (49), and inverse proportion ratio (51):

$$m = k_m/r \quad \wedge \quad r = ct \quad \Rightarrow \quad m(ct)^2 = (k_m/r)r^2 = k_m r. \tag{105}$$

$$m(ct)^2 = k_m r \quad \Rightarrow \quad E \stackrel{\text{def}}{=} mc^2 = k_m r/t^2. \tag{106}$$

$$E = mc^2 = k_m r/t^2 \;\wedge\; r/t = c \;\Rightarrow\; E = mc^2 = k_m c/t. \tag{107}$$

The value of k_m, will be derived from vacuum permeability constant, μ_0 (84), fine structure constant, α (103), and elementary (electron) charge, q_e.?

$$q_p = \sqrt{q_e^2/\alpha} \approx 1.8755460376 \cdot 10^{-18}\ C. \tag{108}$$

$$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2$$

$$\Rightarrow \quad k_m = \mu_0 q_p^2/4\pi \approx 3.51767294 \cdot 10^{-43}\ kg\ m. \tag{109}$$

$$k_m q_p^2 \approx 1.054571783 \cdot 10^{-34}\ k\ m^2\ s^{-1} \approx \hbar. \tag{110}$$

$$E = mc^2 = k_m c/t \ \wedge \ k_m c = \hbar \ \Rightarrow \ E = mc^2 = \hbar/t\ kg\ m^2\ s^{-2} = \hbar/t\ J. \tag{111}$$

$$E = \hbar/t\ kg\ m^2\ s^{-2} \quad \wedge \quad \omega = (2\pi/t)\ radian\ s^{-1} \quad \Rightarrow$$

$$E_\omega = \hbar(2\pi/t) = \hbar\omega\ kg\ m^2\ radian\ s^{-2}. \tag{112}$$

$$E_\omega = \hbar/t\ kg\ m^2\ radian\ s^{-2} \quad \wedge$$

$$f = (1/t)\ cycle\ s^{-1} = ((2\pi/t)\ radian\ s^{-1})/(2\pi\ radian\ cycle^{-1})$$

$$\Rightarrow \quad E_f = \hbar\omega(2\pi/2\pi) = 2\pi\hbar f = hf\ kg\ m^2\ cycle\ s^{-2} = hf\ kg\ m^2\ Hz\ s^{-1}. \tag{113}$$

L. Values of the unit ratios and Planck units

Values of the direct proportion ratios, c_t, c_m, and c_q:

$$c_t = c = 2.99792458 \cdot 10^8\ m\ s^{-1}. \tag{114}$$

$$G = c_m c_t^2 \quad \wedge \quad G \approx 6.67430(15) \cdot 10^{-11}\ m^3\ kg^{-1}\ s^{-2} \quad \Rightarrow$$

$$c_m = G/c_t^2 \approx 7.426162 \cdot 10^{-28}\ m\ kg^{-1}. \tag{115}$$

$$k_e = c_q^2 c_t^2/c_m \quad \wedge \quad k_e \approx 8.9875517923(13) \cdot 10^9 Nm^2\ C^{-2} \quad \Rightarrow$$

$$c_q = \sqrt{k_e c_m/c_t^2} \approx 8.6175182 \cdot 10^{-18} m\ C^{-1}. \tag{116}$$

From equation 111, the values of the inverse proportion ratios, k_t, k_m, and k_q:

$$\hbar \approx .054571783 \cdot 10^{-34}\ kg\ m^2\ s^{-1} \ \wedge \ k_m c = \hbar \ \wedge \ c = 299792458\ m\ s^{-1} \ \Rightarrow$$

$$k_m = \hbar/c \approx 3.51767294 \cdot 10^{-43}\ kg\ m. \tag{117}$$

$$r_p = \sqrt{r_p^2} = \sqrt{k_m c_m} \approx 1.616255 \cdot 10^{-35}\ m. \quad \Rightarrow \tag{118}$$

$$k_t = r_p^2/c_t \approx 8.71363 \cdot 10^{-79}\ s\ m. \tag{119}$$

$$k_q = r_p^2/c_q \approx 3.031361 \cdot 10^{-53}\ C\ m. \tag{120}$$

Values of the Planck units, r_p, t_p, m_p, and q_p::

$$r_p = \sqrt{c_t k_t} = \sqrt{c_m k_m} = \sqrt{c_q k_q} \approx 1.616255 \cdot 10^{-35}\ m. \tag{121}$$

$$t_p = r_p/c_t \approx 5.391246 \cdot 10^{-44}\ s. \tag{122}$$

$$m_p = r_p/c_m \approx 2.176434 \cdot 10^{-8}\ kg. \tag{123}$$

$$q_p = r_p/c_q \approx 1.875546 \cdot 10^{-18}\ C. \tag{124}$$

M. Compton wavelength, λ

[12,16] From equations 51 and 111:

$$rm = k_m \quad \wedge \quad k_m c = \hbar \quad \Rightarrow \quad \lambda = 2\pi r = \frac{2\pi k_m}{m} = \frac{2\pi k_m}{m}\frac{c}{c} = \frac{2\pi\hbar}{mc} = \frac{h}{mc}. \tag{125}$$

N. The classic electron radius, r_e

From the ratio-derived Coulomb's charge force equation 75 and the unit ratios 49:

$$F - \frac{m_e r_e}{t^2} = \frac{k_e q_e^2}{r^2} \quad \Rightarrow \quad m_e r_e = k_e q_e^2 \frac{t^2}{r^2}. \tag{126}$$

$$m_e r_e = k_e q_e^2 \frac{t^2}{r^2} \quad \wedge \quad \frac{r}{t} = c \quad \Rightarrow \quad m_e r_e = \frac{k_e q_e^2}{c^2}. \tag{127}$$

$$m_e r_e = \frac{k_e q_e^2}{c^2} \quad \Rightarrow \quad r_e = \frac{k_e q_e^2}{m_e c^2}. \tag{128}$$

$$r_e = \frac{k_e q_e^2}{m_e c^2} \quad \wedge \quad \varepsilon_0 \overset{\text{def}}{=} \frac{1}{4\pi k_e} \quad \Rightarrow \quad r_e = \frac{q_e^2}{4\pi\varepsilon_0 m_e c^2} \approx 2.81794023 \cdot 10^{-15}\ m, \tag{129}$$

which is classic electron radius definition.[12,16]

O. The Rydberg constant, R_∞

The Rydberg constant, R_∞, is a function of the ratio of the electron's static field forces to the forces of a wave caused by the motion of an electron, which are the fine structure couplings, α (103) and α_G (104). Starting with the ratio-derived classic electron radius (129), rearrange the quantities so that particle quantities are in the numerator and the wave quantities are in the denominator:

$$\frac{r_e}{r_p} = \frac{m_p}{m_e}\frac{q_e^2}{q_p^2} \quad \Rightarrow \quad \frac{r_e}{r_p}\left(\frac{\alpha_G \alpha}{4\pi r_p}\right) = \frac{m_p}{m_e}\frac{q_e^2}{q_p^2}\left(\frac{\alpha_G \alpha}{4\pi r_p}\right) = \frac{}{4\pi r_p}\frac{m_e q_e^4}{m_p q_p^4} = \frac{1}{\lambda} = R_\infty. \tag{130}$$

P. Schrödenger's wave equation

Start with the previous unit ratio-derived "base" Planck-Einstein relation 111, where $V(r,t)$ is the potential energy in a volum. And multiply the kinetic energy component by mc/mc:

$$mc^2 = \hbar/t \quad \Rightarrow \quad \exists\, V(r,t) : \ \hbar/t = \hbar/2t + V(r,t). \tag{131}$$

$$\hbar/t = h/2t + V(r,t) \quad \Rightarrow \quad \hbar/t = \hbar mc/2mct + V(r,t). \tag{132}$$

And from the distance-to-time (speed of light) ratio (49):

$$\hbar/t = \hbar mc/2mct + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar mc^2/2mcr + V(r,t). \tag{133}$$

$$\hbar/t = \hbar mc^2/2mcr + V(r,t) \ \wedge \ \hbar/t = mc^2 \quad \Rightarrow \quad \hbar/t = \hbar^2/2mcrt + V(r,t). \tag{134}$$

$$\hbar/t = \hbar^2/2mcrt + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar^2/2mr^2 + V(r,t). \tag{135}$$

Multiply both sides of equation 135 by a probability density function, $\Psi(r,t)$.

$$\hbar/t = \hbar^2/2mr^2 + V(r,t) \quad \Rightarrow \quad (\hbar/t)\Psi(r,t) = (\hbar^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t). \tag{136}$$

$$\begin{gathered}(\hbar/t)\Psi(r,t) = ((\hbar)^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t) \quad \wedge \\ \forall\ \Psi(r,t) : \ \partial^2\Psi(r,t)/\partial r^2 = (-1/r^2)\Psi(r,t) \quad \wedge \quad \partial\Psi(r,t)/\partial t = (i\ 1/t)\Psi(r,t) \\ \Rightarrow \quad i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t),\end{gathered} \tag{137}$$

which is the one-dimensional position-space Schrödenger's equation.[6,16]

$$\begin{gathered}i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t) \quad \wedge \quad |\vec{\mathbf{r}}| = r \\ \Rightarrow \quad \exists\, \vec{\mathbf{r}} : \ i\hbar\partial\Psi(\vec{\mathbf{r}},t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(\vec{\mathbf{r}},t)/\partial \vec{\mathbf{r}}^2 + V(\vec{\mathbf{r}},t)\Psi(\vec{\mathbf{r}},t),\end{gathered} \tag{138}$$

which is the 3-dimensional position-space Schrödenger's equation.[6,16]

Q. Dirac's wave equation

Start with the previous unit ratio-derived "base" Planck-Einstein relation 111:

$$\hbar/t = mc^2 \quad \Leftrightarrow \quad \hbar/2t + \hbar/2t. = mc^2 \quad \Leftrightarrow \quad \hbar/2t = -\hbar/2t. + mc^2. \tag{139}$$

$$\hbar/2t = -\hbar/2t. + mc^2 \ \wedge \ qc_q/r = qc_q/ct = 1 \ \Rightarrow \ \hbar qc_q/2ct^2 = -qc_q/2rt. + mc^2. \tag{140}$$

$$\hbar qc_q/2ct^2 = -qc_q/2rt. + mc^2 \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar qc_q/2ct^2 = -qcc_q/2r^2. + mc^2. \tag{141}$$

$$\hbar qc_q/2ct^2 = -qcc_q/2r^2. + mc^2 \quad \Rightarrow \quad \hbar(qc_q/2ct^2)\boldsymbol{\psi} = (-qcc_q/2r^2)\boldsymbol{\psi} + mc^2\boldsymbol{\psi}. \tag{142}$$

$$\psi = e^{-iqc_q((1/2ct)-(1/2r))} \ \Rightarrow \ \partial\boldsymbol{\psi}/\partial t = (-iqc_q/2ct^2)\boldsymbol{\psi} \ \wedge \ \nabla\boldsymbol{\psi} = (iqc_q/2r^2)\boldsymbol{\psi}. \tag{143}$$

$$\partial\boldsymbol{\psi}/\partial t = (-iqc_q/2ct^2)\boldsymbol{\psi} \quad \wedge \quad \nabla\boldsymbol{\psi} = (iqc_q/2r^2)\boldsymbol{\psi} \quad \wedge$$
$$\hbar(qc_q/2ct^2)\boldsymbol{\psi} = (-qcc_q/2r^2)\boldsymbol{\psi} + mc^2\boldsymbol{\psi} \quad \Rightarrow \quad i\hbar\partial\boldsymbol{\psi}/\partial t = -i\hbar c\nabla\boldsymbol{\psi} + mc^2\boldsymbol{\psi}. \tag{144}$$

$$\forall\, y = ax + b \ \exists\, f, \boldsymbol{\alpha}, \beta : \ yf(\boldsymbol{\psi}) = \boldsymbol{\alpha} \cdot ax\boldsymbol{\psi} + b\beta\boldsymbol{\psi} \quad \wedge \quad i\hbar\partial\boldsymbol{\psi}/\partial t = -i\hbar c\nabla\boldsymbol{\psi} + mc^2\boldsymbol{\psi}$$
$$\Rightarrow \quad \exists\, f, \boldsymbol{\alpha}, \beta : \ i\hbar\partial\boldsymbol{\psi}/\partial t = i\hbar c\boldsymbol{\alpha} \cdot -\nabla\boldsymbol{\psi} + mc^2\beta\boldsymbol{\psi}, \tag{145}$$

which is the relativistic, free particle Dirac equation, where $\beta = \gamma^0$, $\boldsymbol{\alpha} = \gamma^0\gamma^k$, $k \in \{1, 2, 3\}$, and each γ a matrix.[15]

R. Total of a type

Applying both the direct (49) and inverse proportion ratios (51) to the Euclidean distance:

$$r = \sqrt{r_1^2 + r_2^2}, \quad r_1 = c_\tau\tau, \quad r_2 = k_\tau/\tau, \quad \tau \in \{t, m, q\}$$
$$\Rightarrow \quad r = \sqrt{(c_\tau\tau)^2 + (k_\tau/\tau)^2} \quad \Leftrightarrow \quad \tau = \sqrt{(r/c_\tau)^2 + (k_\tau/r)^2}. \tag{146}$$

S. Quantum extension to Schwarzschild's metric

The simplest way to demonstrate how to add quantum physics to general relativity is by extending Schwarzschild's gravitational time dilation equation and black hole metric. Apply the total of a type equation 146 to the derivation of Schwarzschild's time dilation and metric

(64):

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-(v^2/c^2)(r/r)} \quad \wedge \quad r = \sqrt{(c_m m)^2 + (k_m/m)^2} = Q_m$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-Q_m v^2/rc^2}. \tag{147}$$

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-Q_m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv^2_{escape}$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m v^2_{escape}/rc^2}. \tag{148}$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m v^2_{escape}/rc^2}. \wedge \ v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m c^2/rc^2} = \sqrt{1-2Q_m/r}. \tag{149}$$

Combining equation 149 with equation 59 yields Schwarzschild's gravitational time dilation with a quantum mass effect:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m/r} \quad \wedge \quad t' = t\sqrt{1-(v^2/c^2)} \quad \Rightarrow \quad t' = t\sqrt{1-2Q_m/r}. \tag{150}$$

Schwarzschild defined the black hole event horizon radius, $\alpha \overset{\text{def}}{=} 2Gm/c^2$. The radius with the quantum extension is $\alpha \overset{\text{def}}{=} 2Q_m$. At this point the exact same equations 69 through 72 yield what looks like the same Schwarzschild black hole metric.

T. Quantum extension to Newton's gravity force

The quantum mass effect is easier to understand in the context Newton's gravity equation than in general relativity, because the metric equations and solutions in the EFEs are much more complex. From equations 146 and 49:

$$m/\sqrt{(r/c_m)^2 + (k_m/r)^2} = 1 \quad \wedge \quad r^2/(ct)^2 = 1 \quad \Rightarrow \quad r^2/(ct)^2 = m/\sqrt{(r/c_m)^2 + (k_m/r)^2}$$
$$\Rightarrow \quad r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2}. \tag{151}$$

$$r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2} \quad \Rightarrow \quad (m/r)(r^2/t^2) = (m/r)(c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2})$$
$$\Rightarrow \quad F \overset{\text{def}}{=} mr/t^2 = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{152}$$

$$F = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2} \quad \wedge \quad \forall\, m \in \mathbb{R},\ \exists\, m_1, m_2 \in \mathbb{R} :\ m_1 m_2 = m^2$$
$$\Rightarrow \quad F = c^2 m_1 m_2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{153}$$

U. Quantum extension to Coulomb's charge force

$$q^2/((r/c_q)^2 + (k_q/r)^2) = 1 \;\wedge\; r^2/(ct)^2 = 1 \;\Rightarrow\; r^2/(ct)^2 = q^2/((r/c_q)^2 + (k_q/r)^2)$$
$$\Rightarrow \quad r^2/t^2 = c^2q^2/((r/c_q)^2 + (k_q/r)^2). \tag{154}$$
$$r^2/t^2 = c^2q^2/((r/c_q)^2 + (k_q/r)^2) \quad \wedge \quad r = c_m m$$
$$\Rightarrow \quad F \overset{\text{def}}{=} mr/t^2 = (c^2/c_m)q^2/((r/c_q)^2 + (k_q/r)^2). \tag{155}$$
$$\forall\, q \in \mathbb{R} :\ \exists\, q_1, q_2 \in \mathbb{R} :\ q_1q_2 = q^2 \quad \wedge \quad F = (c^2/c_m)q^2/((r/c_q)^2 + (k_q/r)^2)$$
$$\Rightarrow \quad F = (c^2/c_m)q^2/((r/c_q)^2 + (k_q/r)^2). \tag{156}$$

VI. INSIGHTS AND IMPLICATIONS

1. A distance measure, d, is an inverse (bijective) function of a volume, v, where v is the sum of subset volumes (Euclidean or non-Euclidean), $g(s_{i,1}, \cdots, s_{i,n})$. Notionally:

$$\forall\, n \in \mathbb{N},\ \ f, g :\ d \geq 0,\ \ d = f^{-1}(v) = f^{-1}(\textstyle\sum_{i=1}^{m} g(s_{i,1}, \cdots, s_{i,n})) : \tag{157}$$

   (a) shows the intimate relation between distance and volume that definitions, like inner product space and metric space, ignore.[2,3]

   (b) is a more simple and concise definition of a distance measure, having the metric space properties.[2,3]

2. The left side of the distance sum inequality (III.5),

$$(\textstyle\sum_{i=1}^{m}(a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^{m} a_i^n)^{1/n} + (\sum_{i=1}^{m} b_i^n)^{1/n}, \tag{158}$$

   differs from the left side of Minkowski's sum inequality.[10]

$$(\textstyle\sum_{i=1}^{m}(a_i^n + b_i^n)^{\mathbf{n}})^{1/n} \leq (\sum_{i=1}^{m} a_i^n)^{1/n} + (\sum_{i=1}^{m} b_i^n)^{1/n}. \tag{159}$$

   (a) The two inequalities are only the same where $n = 1$.

   (b) The distance sum inequality (III.5) is a more fundamental inequality because the proof does not require the convexity and Hölder's inequality assumptions required to prove the Minkowski sum inequality.[10]

(c) $\forall\ n > 1,\ v > 0$: The distance sum inequality term, $v_i^n = a_i^n + b_i^n$: $d = v^{1/n} = (\sum_{i=1}^{m} v_i^n)^{1/n}$, is the Minkowski distance, which makes the distance sum inequality directly related to geometry. But the Minkowski sum inequality term, $d = v^{1/n} = (\sum_{i=1}^{m}((v_i^n)^{\mathbf{n}}))^{1/n} = (\sum_{i=1}^{m} v_i^{\mathbf{n^2}})^{1/n}$, is *not* a Minkowski distance.

(d) The distance sum inequality might be useful for comparing distances between concepts in machine learning.

3. *Combinatorics.* The number of ordered combinations (n-tuples) of the elements of disjoint sets was proven to imply the Euclidean volume equation (II.1). And the notion of distance was derived as the inverse function of a volume. The volume and distance equations were derived without relying on the geometric primitives and relations in Euclidean geometry[17,18], axiomatic geometry[19–23], trigonometry[24,25], calculus[24,26,27], and vector analysis.[5]

4. *Combinatorics.* The commutative property of the operations defining volume allows n number of domain values to be sequenced in $n!$ number of permutations. But strict linear sequence limits the set of domain elements to $n \leq$ elements (IV.5). Other elements must have different types (are elements of different sets).

   (a) If there are more than 3 distance and 3 non-distance dimensions $\subseteq \mathbb{R}$, then the additional dimensions $\subseteq \mathbb{R}$ are ordered, where each dimension has an intrinsic property having a fixed position in strict totally ordered set.

   (b) If a set of quantum states is unordered, then there are at most 3 states of the same type.

5. A discrete (point) valued state has measure 0 (zero-length interval). The ratio of a time, distance, mass, or charge interval length to zero is undefined, which is the reason discrete, state values exist independent of distance (position), time, mass, and charge. The value from a set of unordered, discrete state values is undetermined with respect to space and time until observed.

6. For each unit interval length, r_p, of a 3-dimensional distance, there are unit interval lengths of 3 other types of dimensions, forming 3 direct proportion unit ratios (49): $r = (r_p/t_p)t = (r_p/m_p)m = (r_p/q_p)q$ and 3 inverse proportion ratios (51): $r = (r_p t_p)/t = (r_p m_p)/m = (r_p q_p)/q$, where r_p, t_p, m_p, and q_p are the Planck units (121).

7. In this article, the following constants are derived directly from the ratios and Planck units: gravity, $G = c_m c_t^2 = (r_p/m_p)(r_p/t_p)^2 = r_p^3/m_p t_p^2$ (54), reduced Planck $\hbar = k_m c_t = (r_p m_p)(r_p/t_p) = r_p^2 m_p/t_p$ (111), charge, $k_e = c_q^2 c_t^2/c_m$ (76), vacuum permittivity $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$ (80), and vacuum permeability $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m$ $= 4\pi r_p m_p/q_p^2 = 4\pi k_m/q_p^2$ (84).

   And the quantum extensions to: Schwarzschild's gravitational time dilation (149), Newton's gravity force (153), and Coulomb's charge force (156), show that the constants G, $\kappa = 8\pi G/c^4$, k_e, ε_0 and μ_0 do not exist where the quantum effects are measurable.

   Therefore, *none* of the following constants: G, κ, k_e, ε_0, and μ_0 are fundamental constants because 1) they were all shown to be derived solely from the ratios and Planck units and 2) G, κ, k_e, ε_0, and mu_0 are not accurate where the quantum effects are measurable.

8. An empirical law *describes* the relations between the variables in an equation. Deriving empirical laws from the ratios *explains* the reasons for relations between the variables in an equation. Further, all the derivations of the physics equations from the ratios are much shorter and simpler than other derivations, which shows that the ratios are an important tool for scientists and engineers.

9. c_t, c_m, and c_q are the *maximum* ratios: $r = \sqrt{r_1^2 + r_2^2 + r_3^2} \Rightarrow r_1,\ r_2,\ r_3 \leq r$ $\Rightarrow \forall\, \tau \in \{t, m, q\} : r_1/\tau,\ r_2/\tau,\ r_3/\tau \leq r/\tau = r_p/\tau_p = c_\tau$. For example, the speeds of gravity and light waves are limited to the speed, c_t.

10. c_t is a component of the constants: $G = c_m c_t^2$, $k_e = c_q^2 c_t^2/c_m$, $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$, and $\hbar = k_m c_t$. The constants, derived in this article, that do not contain c_t are: vacuum permeability: $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m$, the fine structure, $\alpha = q_e^2/q_p^2$, $\alpha_G = m_e^2/m_p^2$, and Rydberg, $R_\infty = m_e q_e^4/4\pi r_p m_p q_p^4$.

11. The complicated NIST/CODATA definition fo the vacuum permeability constant, $\mu_0 = 2h\alpha/cq_e^2$,[28] reduces to the simpler ratio-derived definition of the vacuum permeability constant, $\mu_0 = 4\pi k_e/c_t^2 = 4\pi k_m/q_p^2$ (84).

Using the ratios to reduce the old definition of μ_0 derived in this article:

$$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2. \quad (160)$$

Using the ratios to reduce the CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$:

$$h = 2\pi k_m c_t,\ \ \alpha = q_e^q/q_p^2 \ \Rightarrow \ \mu_0 = 2h\alpha/cq_e^2 = 4\pi k_m c_t(q_e^2/q_p^2)/c_t q_e^2 = 4\pi k_m/q_p^2. \quad (161)$$

12. The ratio-based derivation of Lorentz's law here (86) differs from other relativistic derivations, in that equations for the magnetic field, $\mathbf{B}$, and magnetic permeability, μ_0, are derived in the process. The equation, $\mathbf{B} = (2\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^2$, allows a simple derivation of the Faraday's law (91) and the magnetic component of Maxwell's electromagnetic equations (101).

13. Maxwell's derivation of the electric and magnetic field wave equations required: Gauss's electric field law in a vacuum: $\nabla \cdot \mathbf{E} = -\rho/\varepsilon_0$, Gauss's magnetic field law in a vacuum: $\nabla \cdot \mathbf{B} = 0$, Faraday's Law: $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$, and Ampere's Law: $\nabla \times \mathbf{B} = \mu_0\varepsilon_0\partial\mathbf{E}/\partial t$.[12] In contrast, the derivations (96) (101), in this article, did not require any of those laws. And the derivations, in this article, were shorter and simpler.

14. As shown in the subsection deriving the Schwarzschild's gravitational time dilation and black hole metric (64)[13,14] using ratios illustrates a simple way of finding exact solutions to Einstein's field equations.

15. Einstein's relativity equations: 1) assume the Lorentz transformations, 2) assume the laws of physics are same at each local Euclidean-like coordinate point, 3) assume the notion of light, and 4) assume that the speed of light is the same at each coordinate point.[4,5]

    The derivations, in this article, were made without those assumptions (does not even require the notion of light). The unit ratios are ratios of Euclidean distance to time, mass, and charge. Therefore, the unit ratios (like the maximum speed ratio, c_t), and all equations (laws) and constants derived from those ratios and Euclidean distance must be the same at each local Euclidean-like coordinate point.

16. There are 3 Planck-Einstein relations: $E_\omega = \hbar\omega\ kg\ m^2\ radian\ s^{-2}$ (112) and $E_f = hf\ kg\ m^2\ Hz\ s^{-1}$ (113), which are both instances of a base Planck-Einstein relation,

$E = mc^2 = \hbar/t\ kg\ m^2\ s^{-2} = \hbar/t\ J$ (111). That is, the assumption that the value, 1, in the term, $E = \hbar(1/t)$, has units is incorrect.

17. CODATA defines the Planck constant to be an "atomic" constant with a defined empirical value and defines the reduced Planck constant as $\hbar = h/2\pi$.[28] But, it was shown, in this article, that the reduced Planck constant $\hbar = k_m c_t$, where the full Planck constant, $h = 2\pi\hbar$ Therefore, the full and reduced Planck constants are not atomic constants.

18. Derivations of Schrödenger's wave equation (138)[6] and Dirac's wave equation (145)[15] from the base Planck-Einstein relation, which uses the reduced Planck constant, are shorter, simpler, and require fewer assumptions. And derivations of the Planck units from the reduced Planck constant are simpler and shorter.

    Therefore, it would be more logical for CODATA to define the reduced Planck constant as, $\hbar = k_m c_t$ and define the full Planck constant in terms of the reduced Planck constant, $h = 2\pi\hbar$.

19. Each type of fine structure being the ratio of an electron's particle's static field force to the force of a wave caused by the motion of an electron, which reduces to the ratio of the square of the units forces is a more parsimonious explanation that results in simpler definitions (equations).

    (a) The following steps show that the unnecessarily complicated CODATA definition, $\alpha = q_e^2/4\pi\varepsilon_0\hbar c$, reduces to the simpler equation-derived in this article:

    $$\varepsilon_0 = 1/4\pi k_e = 1/(4\pi(c_q^2 c_t^2/c_m) \quad \wedge \quad \hbar = k_m c_t \quad \wedge \quad h = 2\pi\hbar$$
    $$\Rightarrow \quad \varepsilon_0 hc = 2\pi k_m c_t^2/(4\pi(c_q^2/c_m)c_t^2) = k_m/(2(c_q^2/c_m))$$
    $$= m_p r_p/(2((r_p/q_p)^2/(r_p/m_p))) = q_p^2/2. \quad (162)$$

    $$\alpha = q_e^2/2\varepsilon_0 hc \quad \wedge \quad \varepsilon_0 hc = q_p^2/2 \quad \Rightarrow \quad \alpha = q_e^2/2(q_p^2/2) = q_e^2/q_p^2. \quad (163)$$

    (b) The unnecessarily complicated fine structure electron gravity coupling constant reduces to the simpler equation derived in this article:

    $$\alpha_G = Gm_e^2/\hbar c = (c_m c_t^2)m_e^2/(k_m c_t)c_t = (r_p/m_p)m_e^2/m_p r_p = m_e^2/m_p^2. \quad (164)$$

20. The complicated NIST/CODATA definition of the Rydberg constant reduces to the simpler ratio-derived equation (130):

$$R_\infty = \frac{m_e q_e^4}{8\varepsilon_0^2 h^3 c} = \frac{m_e q_e^4}{8(1/4\pi k_e)^2 (2\pi\hbar)^3 c_t} = \frac{m_e q_e^4 (4\pi k_e)^2}{8(2\pi\hbar)^3 c_t} = \frac{m_e q_e^4 (c_q^2 c_t^2/c_m)^2}{4\pi (k_m c_t)^3 c_t}$$
$$= \frac{m_e q_e^4 (c_q^2/c_m)^2}{4\pi k_m^3} = \frac{m_e q_e^4 c_q^4}{4\pi k_m^3 c_m^2} = \frac{m_e q_e^4 (r_p/q_p)^4}{4\pi (r_p m_p)^3 (r_p/m_p)^2} = \frac{}{4\pi r_p} \frac{m_e q_e^4}{m_p q_p^4} = \frac{\sqrt{\alpha_G}\alpha^2}{4\pi r_p}. \quad (165)$$

The Rydberg constant, R_∞, as a function of the ratio of an electron's particle's static field force to the force of a wave caused by the motion of an electron (the fine structure couplings, α (103) and α_G (104).) is a more parsimonious explanation explanation.

21. The quantum extensions to: Schwarzschild's black hole metric (72), Newton's gravity force (153), and Coulomb's charge force (156) make quantifiable predictions:

    (a) For gravity, $\lim_{r\to 0} F = c^2 m_1 m_2 / k_m$, and for charge, $\lim_{r\to 0} F = 0$. Finite maximum gravity and charge forces: 1) eliminates the problem of forces going to infinity as $r \to 0$, 2) might allow radioactivity without the need for a weak force, and 3) finite sloped energy well walls might facilitate quantum tunneling.

    (b) The quantum-extended relativity and classic equations reduce to the non-extended relativity and classic equations and constants, where the distance between 2 masses and between 2 charges is sufficiently large or the masses and charges sufficiently large that the quantum effects are not measurable. *Note* that G, k_e, ε_0, mu_0, and κ (Einstein's constant, which contains G) do not exist where the quantum effects becomes measurable.

    (c) The covariant tensor components, in Einstein's field equations, that had the units $1/distance^2$, will now have the more complex units, $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$, $\tau \in \{t, m\}$.

    (d) $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$ implies that as distance $\to 0$, spacetime curvature peaks at the Planck unit length (distance), r_p (121). The ratio of quantum units, $m_p/r_p^3 \approx 5.15484 8\dot{1}0^{96}\ kg\ m^{-3}$, might indicate a maximum mass density. A finite force (spacetime curvature) and finite mass density might imply that black holes have sizes > 0 (are not singularities). It might allow black hole evaporation. If there was a "big bang," then it might not have originated from a singularity.

REFERENCES

[1]Copyright © 2026 George M. Van Treeck. All rights reserved.

[2]R. R. Goldberg, *Methods of Real Analysis* (John Wiley and Sons, 1976).

[3]W. Rudin, *Principles of Mathematical Analysis* (McGraw Hill Education, 1976).

[4]A. Einstein, *Relativity, The Special and General Theory* (Princeton University Press, 2015).

[5]H. Weyl, *Space-Time-Matter* (Dover Publications Inc, 1952).

[6]E. Schrödenger, *Collected Papers On Wave Mechanics* (Minkowski Institute Press, 2020).

[7]W. Conradie and V. Goranko, *Logic and Discrete Mathematics* (Wiley, 2015).

[8]X. P. N. Guigui, N. Miolane, "Introduction to riemannian geometry and geometric statistics: from basic theory to implementation with geomstats," Foundations and Trends in Machine Learning **16**, 182 (2023), https://inria.hal.science/hal-03766900v2, hal-00644459.

[9]Rocq, "Rocq proof assistant," (2025), https://rocq-prover.org/docs.

[10]H. Minkowski, *Geometrie der Zahlen* (Chelsea, 1953) reprint.

[11]I. Newton, *The Mathematical Principles of Natural Philosophy* (Patristic Publishing, 2019) reprint.

[12]R. Feynman, R. Leighton, and M. Sands, *The Feynman Lectures on Physics* (California Institute of Technology, 2010) https://www.feynmanlectures.caltech.edu/.

[13]K. Schwarzschild, "Über das gravitationsfeld eines massenpunktes nach der einsteinschen theorie," Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften **7**, 198–196 (1916), https://archive.org/stream/sitzungsberichte1916deutsch #page/188/mode/2up.

[14]S. Antoci and A. Loinger, "An english translation of k. schwarzschild's 1916 article: On the gravitational field of a mass point according to einstein's theory," ARXIV.org (1999), https://arxiv.org/abs/physics/9905030.

[15]P. Dirac, *The Principles of Quantum Mechanics* (Snowball Publishing, 1957).

[16]M. Jain, *Quantum Mechanics: A Textbook for Undergraduates* (PHI Learning Private Limited, New Delhi, India, 2011).

[17]D. E. Joyce, "Euclid's elements," (1998), http://aleph0.clarku.edu/ djoyce/java/elements/elements .html.

[18]E. S. Loomis, *The Pythagorean Proposition* (NCTM, 1968).

[19]G. D. Birkhoff, "A set of postulates for plane geometry (based on scale and protractors)," Annals

of Mathematics **33** (1932).

[20]D. Hilbert, *The Foundations of Geometry (2cd Ed)* (Chicago: Open Court, 1980) http://www.gutenberg.org/ebooks/17384.

[21]J. M. Lee, *Axiomatic Geometry* (American Mathematical Society, 2010).

[22]O. Veblen, "A system of axioms for geometry," Trans. Amer. Math. Soc **5**, 343–384 (1904), https://doi.org/10.1090/S0002-9947-1904-1500678-X.

[23]A. Tarski and S. Givant, "Tarski's system of geometry," The Bulletin of Symbolic Logic **5**, 175–214 (1999).

[24]A. Bogomolny, *Pythagorean Theorem* (Interactive Mathematics Miscellany and Puzzles, 2010) http://www.cut-the-knot.org/pythagoras/ CalculusProofCorrectedVersion.shtml.

[25]J. Zimba, "On the possibility of trigonometric proofs of the pythagorean theorem," Forum Geometricorum **9**, 275–278 (2009).

[26]B. C. Berndt, "Ramanujan-100 years old (fashioned) or 100 years new (fangled)?" The Mathematical Intelligencer **10** (1988).

[27]M. Staring, "The pythagorean proposition a proof by means of calculus," Mathematics Magazine **69**, 45–46 (1996).

[28]CODATA, "Physical constant values," (2022), https://physics.nist.gov/cuu/Constants/index.html.